Research Statement

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Markets often fail to price externalities. Pollution, traffic risk, carbon, and the value of ecosystems all carry costs or benefits that no market price reflects. My research measures these externalities and asks who gains and who loses when a policy or a large shock changes them. I combine quasi-experimental methods with Taiwan's administrative data, including the agricultural census, transit records, accident registries, and high-frequency transaction records. The work falls into two programs. One studies the rural and environmental economy. The other studies the urban and regional economy. Both turn a hard-to-see externality into a number that policymakers can act on.

1. Agricultural and Environmental Economics

Farms must adapt to climate change while also helping to cut emissions, and they rely on ecosystems whose value rarely enters any price. Governments now intervene heavily through solar subsidies and carbon border tariffs, yet credible evidence on whether these measures reach their targets is scarce.

Liou et al. (2025a) study rooftop solar on Taiwan's livestock and poultry farms using the 2020 agricultural census. Farms that installed rooftop photovoltaics report higher product sales and use more farm labor. Energy transition and farm income can therefore move together rather than trade off against each other. Lee et al. (2025a) reach the same conclusion for chicken farms. Adopters sell about 5.8% more, and the gain comes from higher production volume rather than higher quality.

Liou et al. (2025b) move from the farm to the border. They build a strategic trade model with green-energy investment and show that the EU carbon border tariff pushes non-EU firms toward cleaner production and reshapes competition among them. Lee et al. (2025b) turn to ecosystems. Greater tree diversity lowers both the mean and the variability of forest-farm revenue, and the effect on variability is larger. Biodiversity therefore stabilizes farm income and carries an economic value beyond its ecological one. Chang et al. (2024) trace the pollution spike around Tomb-Sweeping Day to holiday traffic rather than the burning of ritual paper. This corrects a common misattribution about the source of the externality.

2. Urban, Regional, and Transportation Economics

Moving people generates some of the largest externalities in modern life, including traffic deaths, carbon emissions, and congestion. These effects are hard to identify because transport rules and economic shocks are rarely assigned at random.

Liou et al. (2025c) evaluate the Taipei Metro monthly pass with a difference-in-differences design on station-to-station records. The pass raised ridership by 4.3% and passenger-kilometers by 4.6%, with larger effects on weekends and at higher fares. It lowers net carbon emissions once most of the added trips replace car or motorcycle travel. Liou and Chang (2026a) use a reform in Tainan that removed the hook-turn rule in some townships. A difference-in-discontinuity design shows that accidents fell by 21% and victims by 19%, driven mainly by fewer injuries rather than fewer deaths. Medical spending in the treated areas fell by about 4.7%. Liou and Chang (2026b) study the Dajia Mazu pilgrimage, a multi-day event that moves dense crowds along more than 300 kilometers of public road. Traffic accidents, especially injuries, rise sharply around the event, which gives traffic managers a clear window of elevated risk.

A second set of studies uses the pandemic to see how behavior and regional economies absorb a shock. Chang et al. (2021) provide the first causal estimate of COVID-19 on metro use. Taiwan imposed no lockdown, so the response was voluntary. Each additional confirmed case cut ridership by 1.43%, and the drop was larger at stations near night markets, malls, and colleges. Perceived risk, not mandates, drove the decline. Liou et al. (2024) model how consumers update their beliefs during the pandemic using real-time credit-card data. Spending follows a perfect Bayesian updating pattern. Households spent more on clothing and transportation offline and shifted food spending from offline to online stores. Liou et al. (2026) track every tourist hotel in Taiwan from 2015 to 2024. Revenue and employment recovered unevenly across hotel types and locations, and labor-market rigidity slowed the rebound.

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